Homework 3

Ex. 3.2 (10 pts)

No. The hierarchy would survive, because the premise of the question fails: a device's access time is set by its material *and* its size, and at terabyte scale it is size that dominates.

Every read must physically cross the array: the address propagates through the decoder to a word line, and the data returns along a bit line through a sense amplifier. That time is bounded below by the time needed to cross the array, whose linear dimension grows as $\sqrt{C}$ in the capacity C. Take the densest cell in production, a 6T SRAM bit at $0.021\,\mu\mathrm{m}^2$; the faster high-performance variant is larger at $0.025\,\mu\mathrm{m}^2$, so this keeps the bound conservative. One terabyte is 8×10^{12} cells covering $1.68 \times 10^5\,\mathrm{mm}^2$, about 2.4 whole 300 mm wafers and some 0.41 m on a side. Crossing that and returning takes at least 2.7 ns *in vacuum*, eleven clock cycles at 4 GHz, before a single transistor switches; real on-chip wires are several times slower than c. A terabyte of the fastest material is thus nowhere near as fast as a kilobyte of it.

The penalty is already visible within one technology. L1, L2 and L3 on a modern CPU are all SRAM, on the same die at the same process node, yet they answer in roughly 4, 14 and 40 cycles. Only capacity differs: longer word and bit lines carry more RC delay, deeper decoders add stages, and larger bit-line capacitance makes sense amplifiers resolve more slowly. So the hierarchy is not an artifact of mixing cheap slow memory with expensive fast memory. It reappears with the material held fixed.

Power says the same thing. The fastest cells are the leakiest, six transistors per bit with static current in each and no idle state as cheap as a DRAM capacitor's. A terabyte of them could not be powered or cooled as one block, and spreading it out to cool it pays the propagation delay back again.

The right use of such a device would therefore be to put a small fast array in front of it to absorb the references that locality makes predictable: to build the hierarchy inside it. Cost per bit would stop being the motive; the device's own size–speed trade-off would supply one.